**6.**

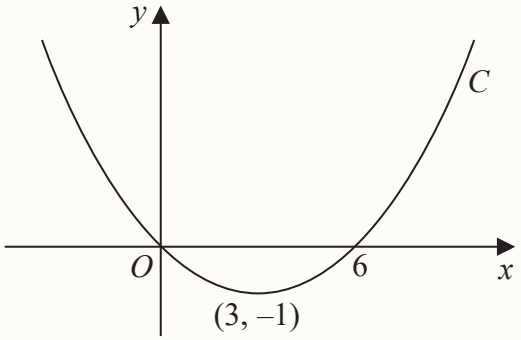



**Figure 1**

Figure 1 shows a sketch of the curve $C$ with equation $y = f(x)$

The curve $C$ passes through the origin and through (6, 0)

The curve $C$ has a minimum at the point (3, –1)

On separate diagrams, sketch the curve with equation

(a) $y = f(2x)$

**(3)**

(b) $y = f(x + p)$, where $p$ is a constant and $0 < p < 3$

**(4)**

On each diagram show the coordinates of any points where the curve intersects the $x$-axis and of any minimum or maximum points.